

Lecture 2 CS 3210

$x := 2$
 $i := 1$

For $i = 1$ to n do

[$x := x \times x$

Print x

$x := 2^{2^n}$

Proof by induction

$$\left(2^{2^n}\right)^2 = 2^{2 \cdot 2^n} = 2^{2^{n+1}}$$

steps in RAM
 $= O(n)$

$$2^{2^0} = 2$$

$$2^{2^{n+1}}$$

How many digits/lits in 2^{2^n} ?

$$2^n + 1 \text{ Lits}$$

The cost of multiplying 2 b-bit numbers is $O(b^2)$ or some function of b depending on algorithm

The unit cost RAM is not a true reflection of the running time

The time to operate on two numbers should be proportional to the "size" of the numbers — not magnitude

$$\left| x_b x_{b-1} \dots x_1 \right| = 2^{\# \text{ bits/digit}}$$

$$\text{Magn.} \leq 2^b$$

$T(n) / S(n)$ is measured in terms
of input size / # input bits

Given an integer m ,
find its factors

Simple algorithm for factorization

Analyse the time $\int$ Try from $k = 2$ to $\sqrt{m}$
If k divides m , then k

running

L

is a factor

else prime

Recurrences / recursive definitions

Given x, n where n is a +ve integer

compute $x^n = \underbrace{x * x * x * \dots * x}_{n \text{ times}}$
 n multiplications

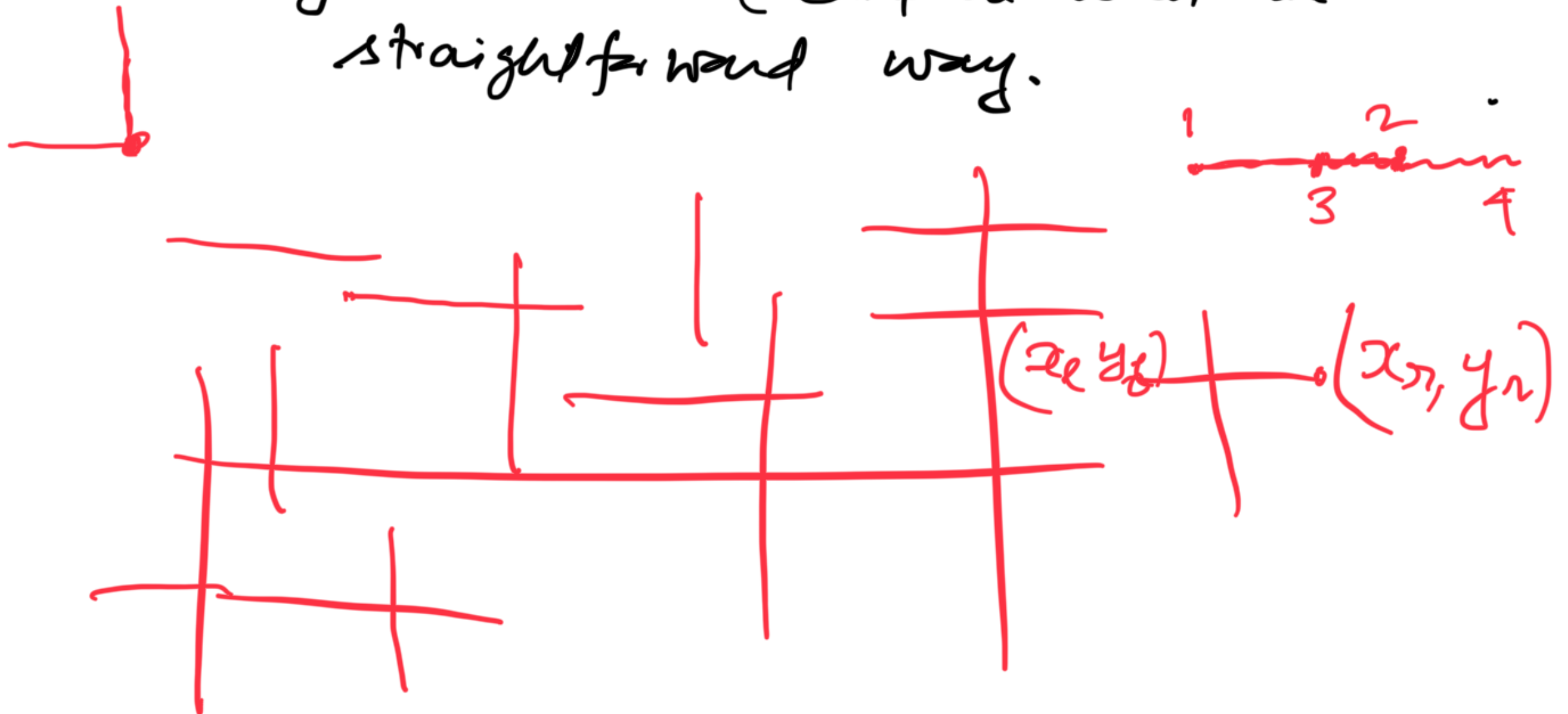
$|x|: b \text{ bits}$ $|x^n| \leq n b$

$\Rightarrow O(n^2 b^2)$ time for ^{each of} n mult.
 $\Rightarrow O(n^3 b^2)$
 $\in \frac{1}{n}$ if $n = 0$
 $\sim n$

$$x^n = \begin{cases} x^2 \times x^2 \times \dots \times x^2 & n: \text{even} \\ x \times x^{\frac{n-1}{2}} \times x^{\frac{n-1}{2}} & \text{otherwise} \end{cases}$$

$O(\log n)$ multiplications

Analyse this and compare with the straight forward way.



Prob: Count/report the number of intersections between a given set S of orthogonal line segments

Algorithm to count/report intersections

For every pair of segments, determine if they intersect

Count/report as required

$T(n)$ is $O(n^2)$ assuming we can solve a pair of segs in constant time.

Worst case complexity

the running time of any algorithm for
a size n input and a size h output
is $\Omega(n+h)$ ↑ not known

$$0 \leq h \leq \Theta(n^2)$$

